AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science & Information Technology

Department of Mathematics

MAT2101: Complex Variables, Laplace and Z-transformations (Sections: All)

Midterm Examination

Total Marks: 40 Time: 2 hours

Instruction: Answer all the questions with the given conditions.

1. Answer **ALL** of the following questions:

(a) Evaluate $\mathcal{L}\left\{e^{-3t}\cos 5t\right\}$,

(b) Evaluate $\mathcal{L}\{(3t\,u(t-5)\}$, where u (t-5) is the unit step function,

(c) Evaluate $\mathcal{L}^{-1}\left\{\frac{5}{(s-4)^2+25}\right\}$.

(d) Evaluate $\text{Im}\left\{\frac{(1-i)^2}{-1+3i}\right\}$.

(e) Find the principal argument of $z=3-i\sqrt{3}$.

2. Answer the following questions:

(a) Given $f(t)=\begin{cases}2t^2, & 0<t<3\\ 1, & 3<t<5\end{cases}$, and $f(t+5)=f(t)$, then

(i) sketch $f(t)$ and,

(ii) find $\mathcal{L}\{f(t)\}$.

(b) Given

$f(t)=\begin{cases}5, & 0<t<2\\ -2, & 2\le t<3\\ 3, & t\ge 3\end{cases}$, then

(i) sketch $f(t)$,

(ii) express it in terms of unit step function and,

(iii) find $\mathcal{L}\{f(t)\}$.

3. Answer the following questions:

(i) Find inverse Laplace of $F(s)=\dfrac{s^2-3}{(s+3)(s^2+4)}$.

(ii) Find inverse Laplace of $F(s)=\dfrac{2s+3}{(s+1)(s-3)^2}$.

4. Solve the following differential equations using Laplace Transformations

(a) $\dot{y}(t) - y(t) = 3\cosh 2t,\ y(0) = 0.$

(b) $\ddot{y}(t) + 7y(t) = 8e^{-t},\ y(0) = \dot{y}(0) = 0.$

(c) $\dot{x} = x + y$

$\dot{y} = 4x + y;\ x(0) = 1, y(0) = 6.$

where, $\dot{y} \equiv \frac{dy(t)}{dt}$ and $\dot{x} \equiv \frac{dx(t)}{dt}$.

5. Answer the following questions:

(a) Describe and sketch the locus represented by

(i) $1 < |z-2-i| \le 2,$ and (ii) $\mathrm{Re}\left\{\dfrac{1}{z}\right\} < \dfrac{1}{2}.$

(b) Find the roots of the complex equation $z^3 - i = 0$ and also locate them in the complex plane.

(c) Let the rectangular region R in z-plane which is bounded by the lines $x = 1, y = 1,$ $x = 3$ and $y = 4.$ Determine the region R' of the w-plane into which R is mapped under the transformation $w = \frac{1}{2}e^{\frac{\pi i}{2}}z + 2.$